Derivatives Mock Final

Updated November 1, 2026

Eight original syllabus-based questions, 100 points, suggested 120 minutes. Questions and writing space precede separate worked solutions, beginner warm-ups and marking guides.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Derivatives mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Price a forward and test an arbitrage (12 points)
2. Separate margin deposits from trading profit (12 points)
3. Set and revalue a plain interest-rate swap (12 points)
4. Use parity and distinguish payoff from profit (12 points)
5. Work backward through an American option tree (12 points)
6. Connect a delta hedge to Black-Scholes (14 points)
7. Scale Brownian shocks and interpret log returns (14 points)
8. Explain what a delta hedge leaves behind (12 points)

Question 1: Price a forward and test an arbitrage

12 points | Syllabus: forward fair value and arbitrage; extends midterm Q1.

A stock is $100. It pays a known $2 dividend in six months. The continuously compounded risk-free rate is 5% and delivery is in one year. Find the fair forward price. A dealer instead quotes $106. Describe the cash-and-carry trade and its terminal profit per share. Assume borrowing/lending at 5%, no fees, and the dividend is certain.

Show your setup, units, calculation and interpretation.

Question 2: Separate margin deposits from trading profit

12 points | Syllabus: futures mechanics; extends midterm Q2-Q3.

You short two futures contracts at 4,000. Each has a $50 multiplier per index point. Total initial margin is $2,000 and total maintenance margin is $1,500. Settlement quotes are 4,003 on day 1 and 4,006 on day 2. Margin earns no interest. Show daily P/L, balances before deposits, any required deposit, cumulative P/L and total deposits. Calls occur strictly below maintenance.

Show your setup, units, calculation and interpretation.

Question 3: Set and revalue a plain interest-rate swap

12 points | Syllabus: swap applications and fair value; extends midterm Q5.

A three-year swap has annual payments, $1m notional and discount factors D1 = 0.96, D2 = 0.91, D3 = 0.85. Calculate the par fixed rate K. In a hypothetical immediate repricing before any floating coupon has fixed, the factors become 0.95, 0.88, 0.80. Find the value to the fixed-rate payer. Assume the floating leg resets at valuation, a single projection/discount curve, and unit accrual periods.

Show your setup, units, calculation and interpretation.

Question 4: Use parity and distinguish payoff from profit

12 points | Syllabus: option basics and strategies.

A non-dividend-paying stock is $100. A European call with strike $100 costs $9 and the strike discount factor is 0.95. Find the matching put price and explain put-call parity as two equal terminal portfolios. Separately, a bull call spread buys a strike-90 call for $15 and sells a strike-110 call for $4. Find its payoff and simple profit at stock prices 80, 100 and 120, plus maximum loss and gain. Ignore financing of premiums for the spread.

Show your setup, units, calculation and interpretation.

Question 5: Work backward through an American option tree

12 points | Syllabus: option modeling and fair value; extension beyond the midterm.

A stock starts at $100. Each of two periods multiplies it by u = 1.2 or d = 0.8. The risk-free growth factor is R = 1.05 each period. Price a strike-$105 American put and the corresponding European put. Show the terminal payoffs, pricing probability, continuation values and early-exercise decision. Assume no dividends and a frictionless tree.

Show your setup, units, calculation and interpretation.

Question 6: Connect a delta hedge to Black-Scholes

14 points | Syllabus: fair-value derivation, Black-Scholes and Greeks.

Assume $dS = \mu S\,dt + \sigma S\,dW$ and a smooth European option $V(S,t)$, no dividends, frictionless continuous hedging and constant r, σ. Using Ito's lemma, explain why a position long V and short V_S shares earns r locally and derive the BSM PDE. Then value a one-year call with $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, given $N(d1) = 0.63683$, $N(d2) = 0.55962$ and $\exp(-0.05) = 0.951229$. State delta and two model limitations.

Show your setup, units, calculation and interpretation.

Question 7: Scale Brownian shocks and interpret log returns

14 points | Syllabus: Wiener processes underlying option models.

For geometric Brownian motion with S0 = 100, μ = 6% annually and σ = 20% annually, use Ito to give the distribution of ln(S_T/S0). At T = 0.25 years, find its mean and standard deviation. Calculate S_T for a standard-normal draw Z = -1. Explain which drift changes under risk-neutral valuation if r = 4% and there are no dividends.

Show your setup, units, calculation and interpretation.

Question 8: Explain what a delta hedge leaves behind

12 points | Syllabus: option Greeks and derivative uses.

A dealer is short 100 calls, each covering 100 shares. Each per-share call has delta 0.60, gamma 0.02 per dollar, and vega $0.15 per one percentage-point volatility increase. How many shares make the position initially delta neutral? Estimate the hedged P/L for a $2 stock rise at unchanged volatility using $0.5\,\Gamma(\Delta S)^2$. Separately estimate P/L for a 3-percentage-point volatility rise at unchanged stock price. Ignore time decay, funding and higher-order terms. Explain the hedge limitations.

Show your setup, units, calculation and interpretation.

Solution 1: Price a forward and test an arbitrage

First, the words

A forward fixes a future purchase price. A known dividend belongs to the stock owner before delivery. Arbitrage means matching cash flows with no net risk under the stated assumptions.

A smaller problem first

Without dividends, buying a $100 stock with a 5% continuously compounded loan creates a one-year cost of 100 exp(0.05) = $105.13.

Hint

Borrow the stock price today; use the dividend to reduce the loan at six months.

Work through it

1. Dividend PV = 2 exp(-0.05 × 0.5) = $1.9506.

2. Fair delivery price = (100 - 1.9506) exp(0.05) = $103.0765.

3. Borrow $100, buy one share, and sell a one-year forward at $106. Net initial cash is zero.

4. At six months use the $2 dividend to reduce the debt. At maturity deliver the share for $106 and repay 100 exp(0.05) - 2 exp(0.025) = $103.0765.

5. Locked terminal profit = $2.9235. This relies on identical timing and frictionless funding.

Mark your reasoning

1. 3: dividend timing and PV

2. 3: fair forward price

3. 4: all trade cash flows

4. 2: profit and assumptions

Hull: forward prices with known income; supplied forward-pricing lecture.

My mark: ____ / 12. My next correction: _____

Solution 2: Separate margin deposits from trading profit

First, the words

Short means you gain when the futures quote falls. Maintenance margin is the trigger; initial margin is the target after a call.

A smaller problem first

A short position of 100 units loses $100 when the quote rises by $1. A deposit of $100 covers collateral but does not undo that loss.

Hint

Daily P/L = -2 × 50 × (new quote - previous quote).

Work through it

1. Total exposure is 100 dollars per point; use the short-position minus sign.

2. Day 1: P/L = -$300; balance = $1,700. No call.

3. Day 2: P/L = -$300; balance before deposit = $1,400, below $1,500.

4. Deposit $600 to restore $2,000. Total deposits, including the original $2,000, are $2,600.

5. Cumulative trading P/L = -$600. Check: deposits $2,600 plus P/L -$600 = final balance $2,000.

Mark your reasoning

1. 2: exposure and sign

2. 3: day 1

3. 3: day 2 and margin call

4. 4: cumulative loss and balance identity

Hull: futures markets and margin; supplied futures-markets lecture.

My mark: ____ / 12. My next correction: _____

Solution 3: Set and revalue a plain interest-rate swap

First, the words

A discount factor is the price today of $1 paid later. A par swap has zero value at inception. A payer swap pays fixed and receives floating.

A smaller problem first

If annual discount factors are 0.96 and 0.90, a two-year par fixed rate is (1 - 0.90)/(0.96 + 0.90).

Hint

At reset, PV of floating coupons is N(1 - D3); PV of fixed coupons is NK times the sum of discount factors.

Work through it

1. Original annuity factor = 0.96 + 0.91 + 0.85 = 2.72.

2. K = (1 - 0.85)/2.72 = 0.05514706, or 5.5147% annually.

3. After repricing, floating coupon PV = 1,000,000 × (1 - 0.80) = $200,000.

4. Fixed coupon PV = 1,000,000 × K × 2.63 = $145,036.76.

5. Payer value = floating minus fixed = +$54,963.24. Away from a reset, account for the already-fixed next floating coupon.

Mark your reasoning

1. 3: original annuity and par rate

2. 3: floating PV

3. 3: fixed PV

4. 3: payer sign and reset assumption

Hull: interest-rate swaps; supplied duration/swaps lecture.

My mark: ____ / 12. My next correction: _____

Solution 4: Use parity and distinguish payoff from profit

First, the words

Payoff is the contract cash at expiry. Profit subtracts its initial cost; this question explicitly ignores financing of that cost.

A smaller problem first

A call with strike $100 has payoff $20 at a stock price of $120. If its premium was $8, simple profit is $12.

Hint

Parity: call + discounted strike = put + stock. Spread payoff is the difference of two call payoffs.

Work through it

1. Put = 9 + 100 × 0.95 - 100 = $4.

2. Call plus a bond paying $100 and stock plus put both pay max(stock price, 100) at expiry; their prices must match.

3. Spread premium paid = 15 - 4 = $11. Payoff = max(S - 90, 0) - max(S - 110, 0).

4. At S = 80, 100, 120: payoffs are $0, $10, $20; simple profits are -$11, -$1, +$9.

5. Maximum loss is $11; maximum gain is $9. Profit and payoff are different graphs.

Mark your reasoning

1. 3: put price

2. 3: replication explanation

3. 4: three payoffs and profits

4. 2: maximum gain/loss

Hull: properties of stock options and trading strategies.

My mark: ____ / 12. My next correction: _____

Solution 5: Work backward through an American option tree

First, the words

Risk-neutral probabilities price tradable claims; they need not be forecasts. An American option allows early exercise.

A smaller problem first

At any American put node, compare the discounted continuation value with immediate exercise max(K - S, 0). Keep the larger.

Hint

q = (R - d)/(u - d). The period-1 down node is the important exercise decision.

Work through it

1. q = (1.05 - 0.8)/(1.2 - 0.8) = 0.625; 0 < q < 1.

2. Terminal stock prices are 144, 96, 64. Put payoffs are 0, 9, 41.

3. At S = 120, continuation = (0.625 × 0 + 0.375 × 9)/1.05 = $3.2143; immediate exercise is 0.

4. At S = 80, continuation = (0.625 × 9 + 0.375 × 41)/1.05 = $20; immediate exercise is $25. American exercise wins.

5. American root value = (0.625 × 3.2143 + 0.375 × 25)/1.05 = $10.8418, greater than immediate exercise $5.

6. European root uses $20 at the down node: value = $9.0561. Early-exercise flexibility is worth $1.7857.

Mark your reasoning

1. 2: q

2. 2: terminal payoffs

3. 4: both continuation/exercise comparisons

4. 2: American root

5. 2: European comparison

Hull: binomial trees and American exercise; supplied Binomial_trees.pdf. This adds depth beyond the short app lesson.

My mark: ____ / 12. My next correction: _____

Solution 6: Connect a delta hedge to Black-Scholes

First, the words

Delta is the local change in option value per $1 stock move. A locally riskless hedge removes the Brownian shock, not all future risk without rebalancing.

A smaller problem first

If a call delta is 0.6, buying 0.6 shares offsets the immediate stock sensitivity of one short call.

Hint

Ito gives $dV = (V_t + \mu S V_S + 0.5\sigma^2S^2V_{SS})dt + \sigma S V_S dW$. Subtract $V_S dS$.

Work through it

1. The self-financing hedged change is $dV - V_S dS = (V_t + 0.5\sigma^2S^2V_{SS})dt$; the dW and μ terms cancel.

2. The hedge value is $V - S V_S$. No arbitrage requires its drift to equal $r(V - S V_S)$.

3. Rearrange: $V_t + 0.5\sigma^2S^2V_{SS} + rS V_S - rV = 0$, with terminal condition $V(S,T) = \max(S - K, 0)$ for a call.

4. $C = S N(d1) - K \exp(-rT) N(d2) = 100 \times 0.63683 - 100 \times 0.951229 \times 0.55962 \approx \$10.45$.

5. Delta = $N(d1) = 0.63683$. For one short option on one share, buy that many shares initially and rebalance.

6. Constant volatility omits smiles and volatility changes; continuous frictionless hedging omits costs, gaps and discrete rebalancing. These limitations matter for actual hedging.

Mark your reasoning

1. 4: cancellation and self-financing reasoning

2. 3: PDE and terminal condition

3. 3: numerical call

4. 2: delta

5. 2: two limitations

Hull: Wiener processes, Ito and the Black-Scholes model. Derivation is an original scaffold, not a reproduced exam answer.

My mark: ____ / 14. My next correction: _____

Solution 7: Scale Brownian shocks and interpret log returns

First, the words

A Brownian increment over length Δt is sqrt(Δt) times a standard-normal draw. Drift scales with time; uncertainty scales with its square root.

A smaller problem first

Annual volatility 20% becomes 10% over a quarter: 20% × sqrt(0.25). It is not 5%.

Hint

For f(S) = ln S, f_S = 1/S and f_SS = -1/S². Include the second-order Ito term.

Work through it

1. Ito gives d ln S = (μ - 0.5σ²)dt + σ dW.

2. Thus ln(S_T/S0) = (μ - 0.5σ²)T + σ sqrt(T) Z, with Z standard normal.

3. Quarter-year mean = (0.06 - 0.5 × 0.20²) × 0.25 = 0.01. Standard deviation = 0.20 sqrt(0.25) = 0.10.

4. For Z = -1, log return = 0.01 - 0.10 = -0.09, so S_T = 100 exp(-0.09) = $91.39.

5. Under the risk-neutral measure replace μ by r = 0.04. Log-return mean becomes 0.005; volatility remains 0.10 in this model.

6. The physical drift describes the assumed real-world process; the pricing drift enforces no-arbitrage valuation. They answer different questions.

Mark your reasoning

1. 4: Ito log transformation

2. 3: distribution and time scaling

3. 3: numerical price

4. 2: risk-neutral drift

5. 2: interpretation

Hull: Wiener processes and Ito; Valuation stochastic-process lecture for supporting intuition.

My mark: ____ / 14. My next correction: _____

Solution 8: Explain what a delta hedge leaves behind

First, the words

Gamma measures how delta changes as the stock moves. Vega measures sensitivity to a volatility change. Here vega is quoted per one percentage point of volatility.

A smaller problem first

A share has delta 1 and gamma 0. Shares can neutralize delta but cannot cancel option gamma.

Hint

There are 10,000 option-share units. Short positions reverse the Greeks' signs.

Work through it

1. Option position delta = -10,000 × 0.60 = -6,000. Buy 6,000 shares.

2. Total option gamma = -10,000 × 0.02 = -200 per dollar. The stock adds zero gamma.

3. After canceling the first-order delta term, estimated stock-move P/L = 0.5 × (-200) × 2^2 = -$400.

4. Total vega = -10,000 × 0.15 = -$1,500 per volatility percentage point. A 3-point increase gives approximately -$4,500.

5. Delta neutrality is local and temporary. Gamma changes delta, volatility changes option prices, and discrete hedging/costs prevent a perfect continuous hedge.

Mark your reasoning

1. 3: multiplier and share hedge

2. 3: gamma P/L

3. 3: vega units and P/L

4. 3: limitations

Hull: Greeks and hedging. Supplied syllabus lists Greeks in the final option block.

My mark: ____ / 12. My next correction: _____